

Benchmarking Arabic LLMs on Cultural Reasoning, Knowledge, Readability, and Translation

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1 Research Objectives

This study aims to systematically benchmark five open Arabic (or Arabic-capable) large language models (LLMs) across four complementary tasks: (1) Arabic cultural common-sense reasoning (ArabCulture), (2) multi-domain academic knowledge and reasoning (ArabicMMLU), (3) readability classification of Arabic texts (BAREC), and (4) sentence-level machine translation (FLORES-200, eng $\leftrightarrow$ arb).

These four dimensions were chosen because they cover key aspects of real-world Arabic applications:

- **Commonsense and cultural grounding** (ArabCulture) reflects the ability of models to handle local knowledge, social norms, and context-sensitive reasoning that are essential for dialogue, storytelling, and culturally appropriate responses.
- **Academic knowledge and reasoning** (ArabicMMLU) evaluates broad factual coverage and reasoning capacity across ~ 40 subjects, relevant for education, tutoring, and knowledge-intensive applications.
- **Readability and text difficulty** (BAREC) measures sensitivity to linguistic complexity, which is crucial for educational tools, adaptive learning systems, and audience-specific content generation.
- **Translation quality** (FLORES-200) assesses adequacy and fluency in English $\leftrightarrow$ Arabic translation, directly informing use cases in cross-lingual communication, localization, and international knowledge access.

By combining these benchmarks, we seek to provide a holistic profile of each model’s strengths and weaknesses. The results not only enable fair comparison but also inform application-specific recommendations: for instance, identifying which models are more suitable for Arabic

story generation (where cultural grounding and fluency are critical) versus translation tasks (where literal accuracy and morphological handling matter most).

Models under test:

- `inceptionai/jais-family-6p7b-chat` (Jais-6.7B-Chat)
- `tiiuae/Falcon3-7B-Instruct` (Falcon3-7B-Instruct)
- `Qwen/Qwen2.5-7B-Instruct` (Qwen3-8B-Arabic)
- `ALLaM-AI/ALLaM-7B-Instruct-preview` (ALLaM-7B-Instruct-preview)
- `silma-ai/SILMA-9B-Instruct-v1.0` (SILMA-9B-Instruct-v1.0)

2 Evaluation Design

2.1 ArabCulture: Cultural Common-Sense (Binary Yes/No)

What it measures. Cultural norms and everyday reasoning grounded in 13 Arab countries (3,482 binary items). Each question requires a Yes/No judgment about a culturally grounded statement.

Protocol. Zero-shot; the model is prompted with the question and two options (Yes/No). Outputs are parsed to detect bias or flipping errors, and a small calibration step was applied if the model systematically inverted answers.

Metric. Accuracy (%). Random baseline = 50%.

2.2 ArabCulture Overall Accuracy (%)

Model	Accuracy (%)
Jais-6.7B (<code>inceptionai/jais-family-6p7b-chat</code>)	93.34
Falcon-7B (<code>tiiuae/Falcon3-7B-Instruct</code>)	98.88
Qwen3-8B Arabic (proxy ¹)	98.36
ALLaM-7B (<code>ALLaM-AI/ALLaM-7B-Instruct-preview</code>)	47.79
SILMA-9B (<code>silma-ai/SILMA-9B-Instruct-v1.0</code>)	79.26

Table 1: ArabCulture overall accuracy (binary Yes/No). Higher is better; random baseline=50%.

Interpretation. Above-chance performance indicates grasp of regional cultural norms; note that extremely high scores can stem from “Yes/No” label calibration working very well on this dataset.

¹Qwen value here comes from the Arabic-multitask variant measured earlier; we use it as a proxy for Qwen2.5. Differences may exist between Qwen3-Arabic and Qwen2.5.

2.3 ArabicMMLU: Academic Knowledge (4-way MCQ)

What it measures. ArabicMMLU is the Arabic adaptation of the Massive Multitask Language Understanding benchmark. It covers about 14,500 multiple-choice questions across ~40 subjects, ranging from primary education to university-level exams. Domains include STEM, humanities, social sciences, and language.

Protocol. Each question has four options (A/B/C/D). Models are prompted in a zero-shot setting with the question and answer choices, and must output the correct letter. Accuracy is calculated overall and by subject group.

Metric. Accuracy (%), with random baseline = 25%. Higher accuracy reflects stronger general knowledge and reasoning in Arabic. Performance differences across subjects (e.g., better on humanities, weaker on STEM) reveal model strengths and weaknesses.

2.4 ArabicMMLU Group-level Accuracy (%)

Group	SILMA-9B	Qwen3-8B	Jais-6.7B	Falcon-7B	ALLaM-7B
arabicmmlu	60.93	64.28	56.20	43.76	70.05
Humanities	57.83	62.49	57.99	45.73	74.75
Language	62.45	63.06	54.31	42.77	73.88
Other	65.74	69.08	61.51	49.80	74.03
Social Science	60.93	62.04	55.25	39.70	66.41
STEM	59.94	65.67	52.02	48.11	63.64

Table 2: ArabicMMLU group-level accuracies by model. Higher is better; random baseline=25%.

2.5 BAREC: Readability Classification

What it measures. The Balanced Arabic Readability Evaluation Corpus (BAREC) tests how well models can judge text difficulty. It includes 69,000+ sentences labeled from Level 1 (very simple, preschool) to Level 19 (advanced academic).

Protocols. We evaluate under two setups:

- **Fine-grained (19-class):** Predict the exact level (1–19).
- **Coarse-grained (5-class):** Levels grouped into five bands (elementary, middle school, high school, undergraduate, postgraduate).

Metrics. We report accuracy. Random baselines are ~5% (19-class) and 20% (5-class). Accuracy is strict (exact match), but reveals how well models can detect sentence complexity.

Interpretation. BAREC evaluates sensitivity to linguistic difficulty, not knowledge. Current Arabic LLMs perform only slightly above chance on 19-class, and modestly better on 5-class, showing that readability remains a challenging task.

2.6 BAREC Overall Accuracy by Level (%)

Model	Level-3	Level-5	Level-7	Level-19
Jais-6.7B	36.76	22.71	13.44	4.23
Falcon-7B	35.82	22.04	15.23	0.45
Qwen3-8B	49.86	21.36	5.04	0.44
ALLaM-7B	44.80	24.40	13.41	0.71
SILMA-9B	50.37	20.53	4.54	0.44

Table 3: BAREC accuracies under different granularity levels. Higher is better; random baselines $\approx 5\%$ (19-class) and 20% (5-class).

Interpretation. All models struggle on fine-grained readability (19-class; very low accuracies), while coarse levels (3/5/7) show slight separation. Qwen3/SILMA top Level-3, ALLaM slightly better on Level-5; none approaches human reliability.

2.7 FLORES-200: Machine Translation (eng $\leftrightarrow$ arb)

What it measures. FLORES-200 is a multilingual machine translation benchmark created by Meta, covering 200 languages with professionally translated evaluation sets. We use the English–Arabic pair (eng_Latn $\leftrightarrow$ arb_Arab) on the `devtest` split (1,012 sentence pairs). Sentences are domain-balanced (sourced from Wikipedia and other formal content) and human-translated, making this benchmark a strong test of both adequacy (faithfulness to the source) and fluency (well-formed target language).

Protocol. Models are evaluated in a zero-shot setting. Each English sentence is translated into Arabic (and vice versa). Decoding is performed with `num_beams=1` (no beam search). This ensures fast and deterministic outputs; using larger beams (e.g., `num_beams=4`) typically improves translation quality at the cost of speed.

Metrics. We report BLEU and chrF, two widely used automatic MT metrics:

- **BLEU (Bilingual Evaluation Understudy)** measures n -gram overlap between system output and the reference translation (up to 4-grams). It applies a brevity penalty to discourage very short outputs. Higher BLEU scores indicate closer lexical matching to the reference translation. BLEU rewards literal accuracy but may penalize acceptable paraphrases.
- **chrF (Character F-score)** measures character n -gram overlap (typically 1–6) between system and reference. Unlike BLEU, chrF operates at the character level and is more sensitive to morphology, spelling, and inflection. This makes it particularly suitable for morphologically rich languages like Arabic. Higher chrF reflects better handling of fluency, morphology, and fine-grained word forms.

How to interpret BLEU vs. chrF together.

- **High BLEU + high chrF:** Translations are both faithful (adequate) and fluent (ideal outcome).
- **High BLEU + lower chrF:** Close lexical match to references but with morphological or fluency issues.
- **Lower BLEU + high chrF:** More paraphrased or freer translations that are fluent and natural, but diverge from the reference wording.

About num_beams=1. Setting `num_beams=1` disables beam search, so the model only produces the most likely next token at each step. This is faster and deterministic but may underuse the model’s full potential. Increasing to `num_beams=4` or higher typically improves adequacy and fluency, at the cost of slower decoding.

2.8 FLORES-200 (eng→arb), devtest

Model	BLEU	chrF	Speed (sps)
Jais-6.7B	3.09	16.47	1.24
Falcon-7B	1.39	21.32	19.47
Qwen2.5-7B	5.30	34.35	19.42
ALLaM-7B	20.63	49.06	15.77
SILMA-9B	8.55	30.70	13.91

Table 4: FLORES-200 English→Arabic results with `num_beams=1`. Higher is better for BLEU/chrF; Speed in sentences per second (sps).

Interpretation. With `num_beams=1`, ALLaM leads on eng→arb; Qwen2.5 shows strong chrF despite lower BLEU.

2.9 FLORES-200 (arb→eng), devtest

Model	BLEU	chrF	Speed (sps)
Jais-6.7B	16.20	46.12	1.22
Falcon-7B	13.46	43.15	13.35
Qwen2.5-7B	10.54	46.87	17.27
ALLaM-7B	20.85	38.45	15.65
SILMA-9B	13.43	33.79	16.94

Table 5: FLORES-200 Arabic→English results with `num_beams=1`. Higher is better for BLEU/chrF; Speed in sentences per second (sps).

Interpretation. ALLaM leads by BLEU on arb→eng; Jais/Qwen2.5 show competitive chrF, hinting at token-level adequacy.

3 Discussion and Model Selection

Cultural Reasoning (ArabCulture). Very high accuracies for Falcon/Qwen proxy reflect the dataset’s “Yes/No calibration” sensitivity; SILMA strong; ALLaM lower on this specific setup.

Readability (BAREC). All models remain far from reliable fine-grained readability judgment; at coarse Level-5, ALLaM slightly ahead; Qwen/SILMA better at Level-3.

Translation. With `num_beams=1`, ALLaM is the most robust overall (both directions). Jais/Qwen2.5 exhibit strong chrF in arb→eng. Beam search (`num_beams=4`) is expected to further improve MT quality.

Which model for Arabic story generation? Prefer SILMA-9B (broad Arabic alignment) or ALLaM-7B (good MT & acceptable culture/knowledge), depending on style and latency constraints; consider Qwen2.5 when logic/format control matters.

Which model for translation? ALLaM-7B leads in BLEU both directions under `num_beams=1`; try `num_beams=4` to lift quality further. For arb→eng where chrF is key, Jais/Qwen2.5 are competitive.

4 Limitations and Next Steps

- **ArabCulture: calibration and robustness.** Current binary Yes/No calibration can inflate scores when a model exhibits label polarity bias. Next, we will (i) report per-country and per-topic breakdowns to reveal cultural coverage gaps; (ii) run prompt-robustness checks (paraphrase the question, invert surface forms like “Yes/No” vs. “True/False” and Arabic variants) to test stability; (iii) add confidence calibration (e.g., reliability diagrams and ECE) to quantify overconfidence; and (iv) introduce adversarially negated items to detect shortcut patterns rather than true cultural understanding.
- **BAREC: ordinal awareness and stratification.** Accuracy on 19-class is near chance, suggesting weak ordinal sensitivity. We will (i) adopt ordinal metrics such as quadratic weighted kappa (QWK) and mean absolute error (MAE) over levels; (ii) analyze performance stratified by sentence length, genre, source, and diacritization to separate linguistic from topical difficulty; (iii) explore ordinal regression style prompting and lightweight calibration that penalizes large level jumps; and (iv) probe error asymmetry (over- vs. under-estimation of difficulty) to guide targeted instruction tuning.
- **MT (FLORES-200): decoding and metrics.** We used `num_beams=1` for speed. Next, we will (i) re-run with `num_beams=4` plus length normalization and coverage penalties, (ii) evaluate constrained decoding with terminology lists for named entities and domain terms, (iii) compare additional metrics beyond BLEU and chrF (e.g., COMET) to better capture adequacy and semantic fidelity, and (iv) add out-of-domain

sets (such as WMT-style news or conversational text) to test robustness and domain transfer.

- **ArabicMMLU: consolidated reporting and reasoning.** We will provide a consolidated table with macro and micro averages across approximately 40 subjects, together with subject-level error profiles. We will also test few-shot and chain-of-thought variants to assess whether reasoning-style prompting changes relative strengths, and run answer-consistency checks across paraphrased items to measure stability.
- **Efficiency and reproducibility.** For all tasks, we will document throughput and latency on A100 under unified batching and precision settings, evaluate the impact of key-value cache and Flash-Attention kernels, and release prompt templates and seeds for exact reproducibility. We will also include memory footprints and batch-size ceilings per model to inform deployment choices.